

Hospital Bed Occupancy Forecasting

Data Requirements Specification (DRS)

Document Version

Version: 1.0

Purpose

The purpose of this document is to outline the data requirements for developing a Machine Learning-based Hospital Bed Occupancy Forecasting solution. The requested data will be used to build predictive models that forecast future bed occupancy, department-wise demand, specialty-wise demand, discharge trends, and bed availability.

1. Data Extraction Requirements

Historical Data Period

- Minimum Required: 3 Months
- Recommended: 6–12 Months
- Preferred: 12+ Months

Data Granularity

One record per patient admission and bed occupancy event.

Accepted Formats

- CSV
 - Excel (.xlsx)
 - SQL Database Extract
 - Data Warehouse Export
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2. Mandatory Data Fields

The following fields are required to build the baseline forecasting model.

Field Name	Description	Example
Patient_ID	Unique patient identifier (masked/anonymized)	P100245
Admission_DateTime	Date and time of admission	2026-06-01 10:30:00
Discharge_DateTime	Date and time of discharge	2026-06-05 16:45:00
Department_Name	Department where patient was admitted	ICU
Specialty_Name	Medical specialty	Cardiology
Bed_Type	Bed classification	ICU Bed
Ward_Type	Ward category	ICU
Admission_Type	Emergency/Elective/Referral	Emergency
Payer_Channel	Insurance/Cash/Corporate/etc	Insurance
Bed_Occupied_Flag	Occupancy indicator (1/0)	1
Bed_Allocation_Timestamp	Time when bed was assigned	2026-06-01 11:00:00

3. Recommended Data Fields

The following fields significantly improve prediction accuracy.

Field Name
Admission_Source
Expected_Discharge_Date
Actual_Discharge_Status
Readmission_Flag
Elective_vs_Emergency_Flag
Bed_Transfer_Count
Transfer_From_Department
Transfer_To_Department
Bed_Release_Time
Bed_Cleaning_Time
Bed_Block_Status

4. Clinical Data Fields

These fields enhance ICU demand and critical-care forecasting.

Field Name

Patient_Severity_Level

ICU_Requirement_Flag

Isolation_Requirement

Ventilator_Requirement

Procedure_Surgery_Flag

Surgery_Date

5. Financial & Payer Information

These fields help improve discharge and occupancy prediction.

Field Name

Insurance_Approval_Time

Corporate_Tieup_Name

Billing_Clearance_Time

Estimated_Bill_Amount

6. Staffing & Resource Information

These fields support operational intelligence and workforce planning.

Field Name

Nurse_Patient_Ratio

Shift_Count

Doctor_Availability

7. Calendar & Seasonality Information

These fields help identify temporal occupancy patterns.

Field Name

Day_of_Week

Weekend_Flag

Holiday_Flag

Seasonal_Period

Examples:

- Monsoon Season
 - Flu Season
 - Festival Season
 - Summer Vacation Period
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8. Advanced Data Fields (Future Enhancements)

The following fields are optional but highly valuable for advanced forecasting models.

Field Name

Ambulance_Arrival_Count

Waiting_List_Count

OPD_Conversion_Rate

Average_Discharge_Delay

Patient_Category

9. Data Tables Requested

To facilitate data extraction, the following Hospital Management System (HMS) tables are requested:

1. Admissions Table
2. Discharges Table
3. Bed Allocation Table

4. Bed Master Table
 5. Department Master Table
 6. Billing / Payer Table
 7. Transfers Table
 8. Surgery / OT Schedule Table
 9. Emergency Department Table
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10. Features Generated During Data Engineering

The Machine Learning pipeline will automatically derive additional features such as:

- Length of Stay (LOS)
 - Admission Hour
 - Admission Day
 - Admission Month
 - Week Number
 - Weekend Indicator
 - Holiday Indicator
 - Seasonal Classification
 - Rolling Occupancy Averages
 - Department Occupancy Rate
 - Specialty Occupancy Rate
 - Bed Utilization Rate
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11. Prediction Outputs

The forecasting platform will generate:

Occupancy Forecasting

- Overall Bed Occupancy Forecast
- Department-wise Occupancy Forecast
- Specialty-wise Occupancy Forecast
- Payer-wise Occupancy Forecast

Bed Management Insights

- Predicted Bed Demand
- Bed Availability Forecast

- Bed Utilization Trends
- Length of Stay Trends

Operational Intelligence

- Occupancy Risk Alerts
 - Capacity Planning Recommendations
 - Discharge Probability Analysis
 - Department Load Forecasting
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12. Minimum Dataset Required for POC

If complete extraction is difficult, the following fields are sufficient to build the Proof of Concept (POC):

1. Patient_ID
2. Admission_DateTime
3. Discharge_DateTime
4. Department_Name
5. Specialty_Name
6. Bed_Type
7. Ward_Type
8. Admission_Type
9. Payer_Channel
10. Bed_Occupied_Flag

This minimum dataset enables:

- Overall Occupancy Forecasting
 - Department-wise Forecasting
 - Specialty-wise Forecasting
 - Payer-wise Forecasting
 - Length of Stay Analysis
 - Bed Availability Prediction
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Data Privacy Requirements

- Patient identifiers must be anonymized or masked.
- No personally identifiable information (PII) is required.

- Data should comply with organizational security and healthcare data governance policies.
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Client Deliverable Summary

Requested Dataset:

- Historical operational hospital admission data
- Bed occupancy records
- Discharge records
- Bed allocation details
- Department and specialty information
- Payer/payment channel information

Preferred Historical Duration:

- 6 to 12 months

Preferred Format:

- CSV or Excel

Update Frequency:

- Daily or Weekly